WSe_2:17-angle Heff joint model | completed production results

Q8: four new angles,61points each. No extra point-group averaging.
Seven physical coefficients; pure quadratic kinetic; ONE common mass = 0.35574179 me.
Five cubic Bernstein angle laws and one linear lambda law in normalized z;23global coefficients.
All17angle continuum basis convergence checks passed (top6 change <0.01meV).

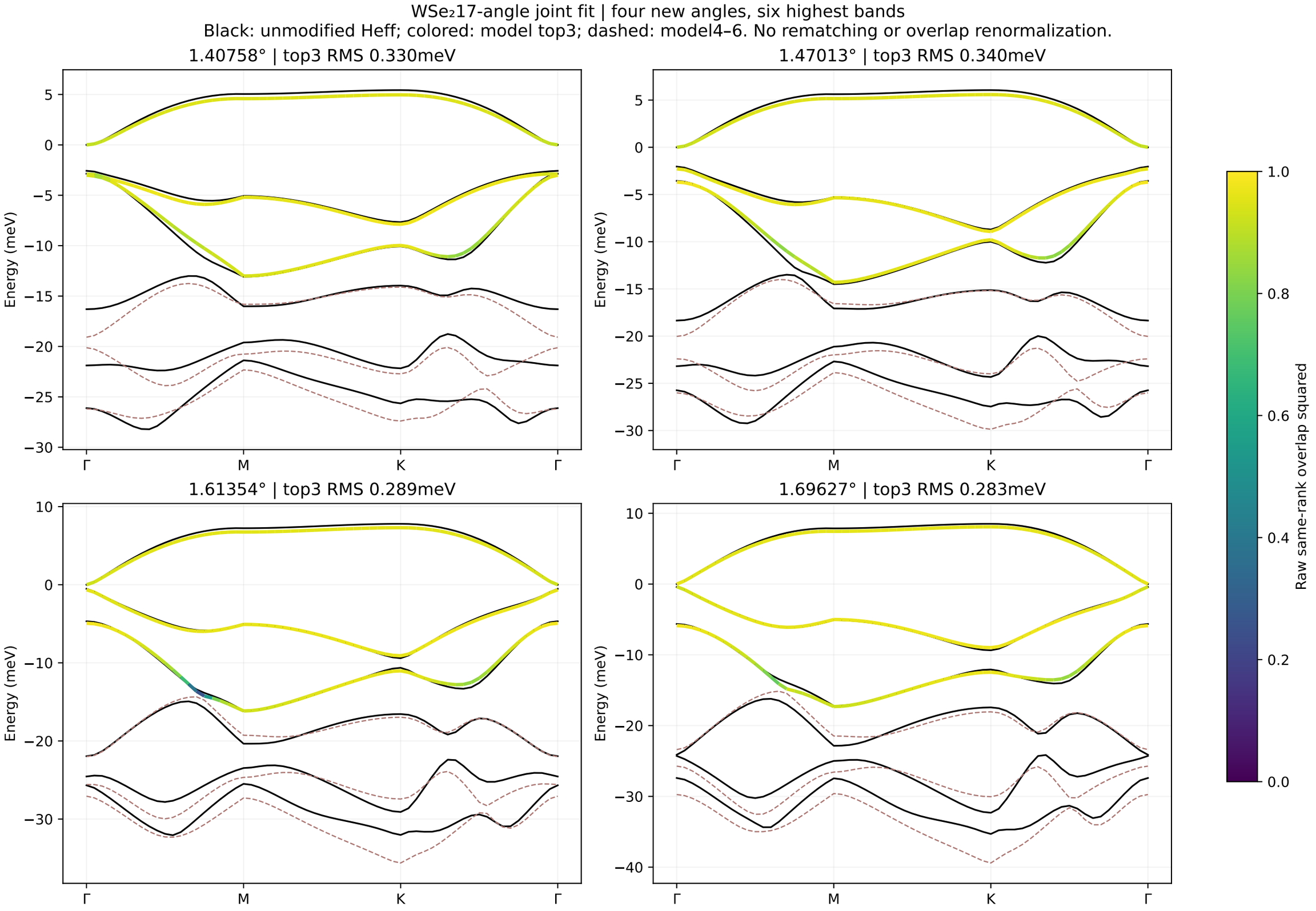
Angle(deg)	Old prediction RMS3	Joint fit RMS3	Gamma $O1^2$ / $O2^2$ / $O3^2$	Model(C1,C2,C3)
1.40758	0.469	0.330	0.906 / 0.963 / 0.989	(0, 0, 0)
1.47013	0.483	0.340	0.910 / 0.965 / 0.988	(0, 0, 0)
1.61354	0.461	0.289	0.916 / 0.970 / 0.985	(0, 2, -2)
1.69627	0.320	0.283	0.973 / 0.919 / 0.984	(1, 1, 1)

Energy RMS values are meV, over the full61point path excluding repeated Gamma.

Model Gamma crossings: bands1/2 at1.66627deg; bands2/3 at1.40192deg.
These are model predictions; no full-BZ Heff Chern validation is implied.
BW/gaps: all88displayed angles use the same60x60 full-BZ mesh.

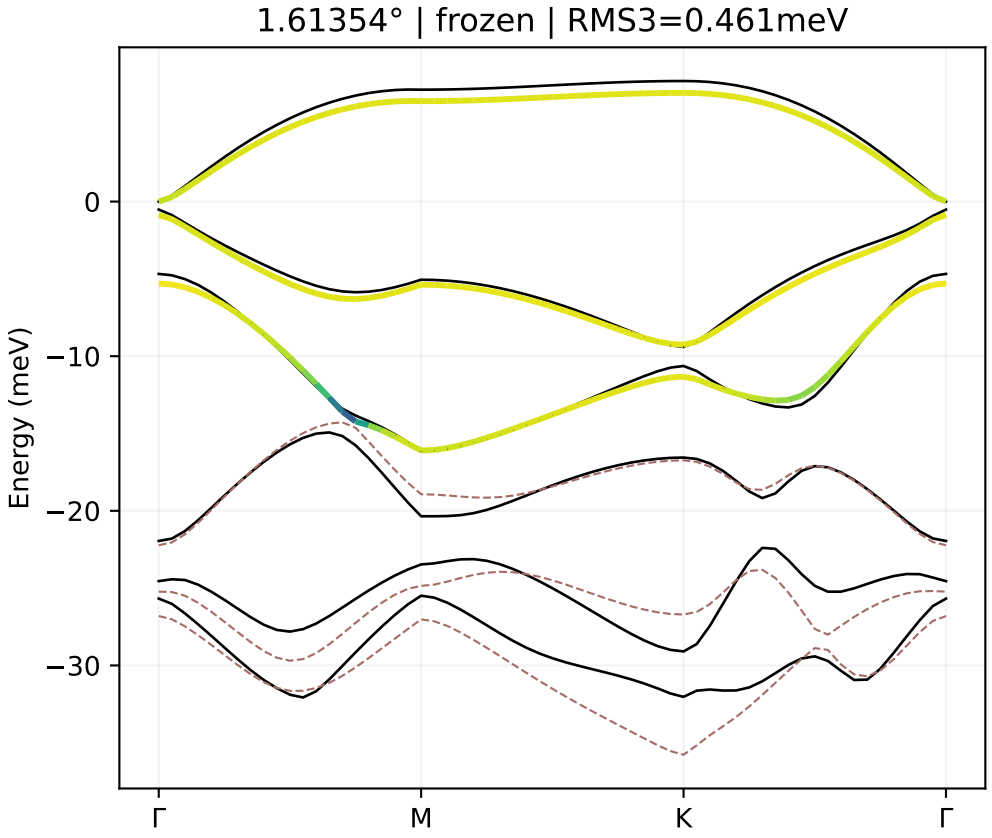
Remaining limitations:
-5.09deg: top3 RMS2.647meV; M-band3 raw overlap² about0.440.
-Local3/4-band exchanges remain visible; F23/F34 listed separately.
-Below1.202855deg is parameter extrapolation; no new endpoint-deletion fits this round.

Numerics: gpuh2002; model job584524 COMPLETED0:0; observables job584847 COMPLETED0:0.
Runtime: /data/home/zy/mambaforge/envs/moirekp. Details and source hashes: REPORT_ZH.md.

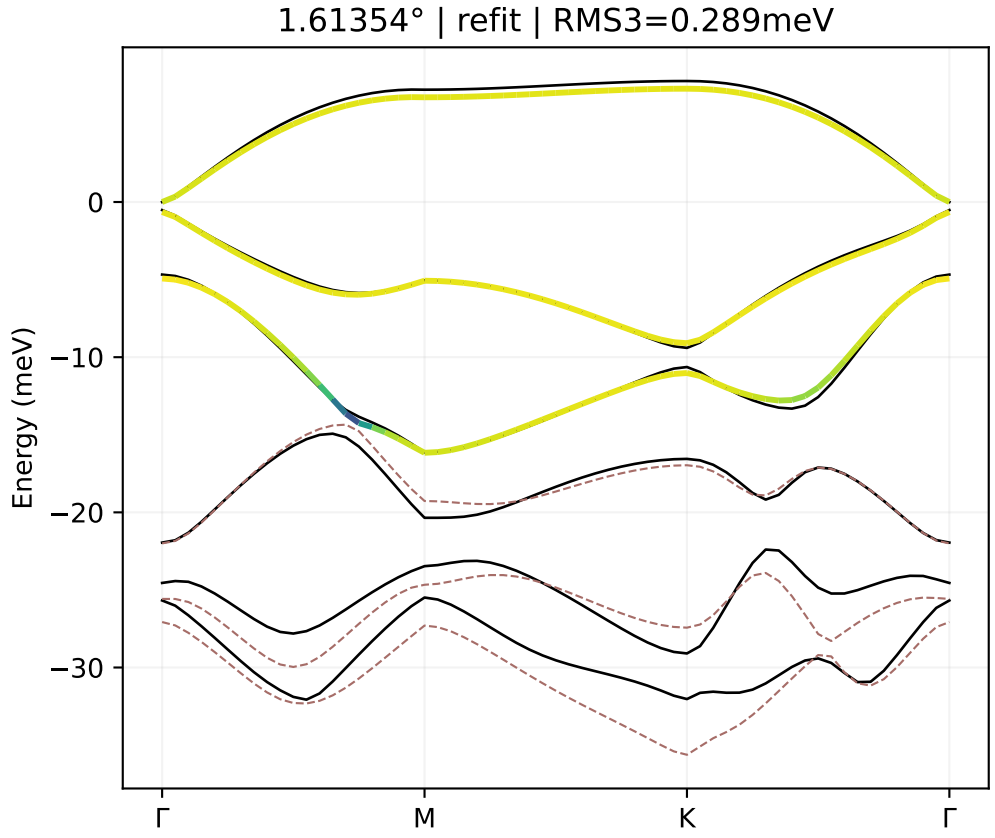

WSe₂17-angle joint fit | four new angles, six highest bands
Black: unmodified Heff; colored: model top3; dashed: model4–6. No rematching or overlap renormalization.
1.40758° | top3 RMS 0.330meV
1.47013° | top3 RMS 0.340meV
1.61354° | top3 RMS 0.289meV
1.69627° | top3 RMS 0.283meV
Energy (meV)
Γ
M
K
Γ
Raw same-rank overlap squared

New WSe_2 angles | black: Heff; colored: model top3; dashed: model4–6
Left: previous13-angle prediction. Right:17-angle joint refit. Accepted constrained update.

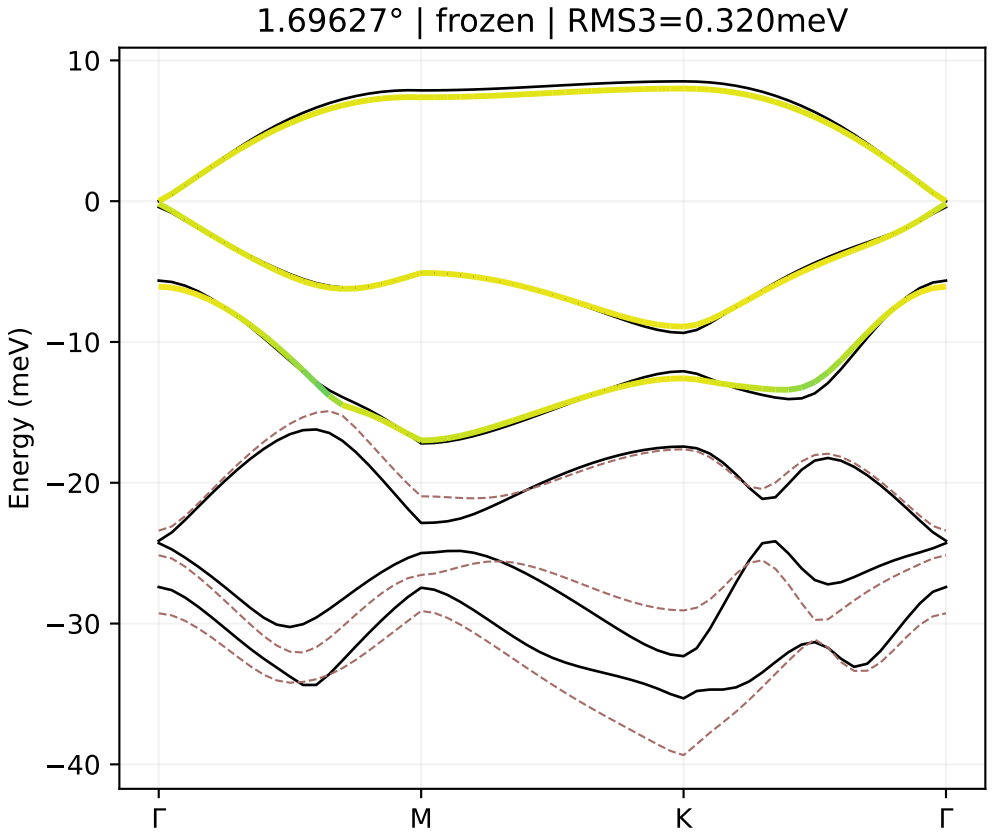
1.61354° | frozen | RMS3=0.461meV



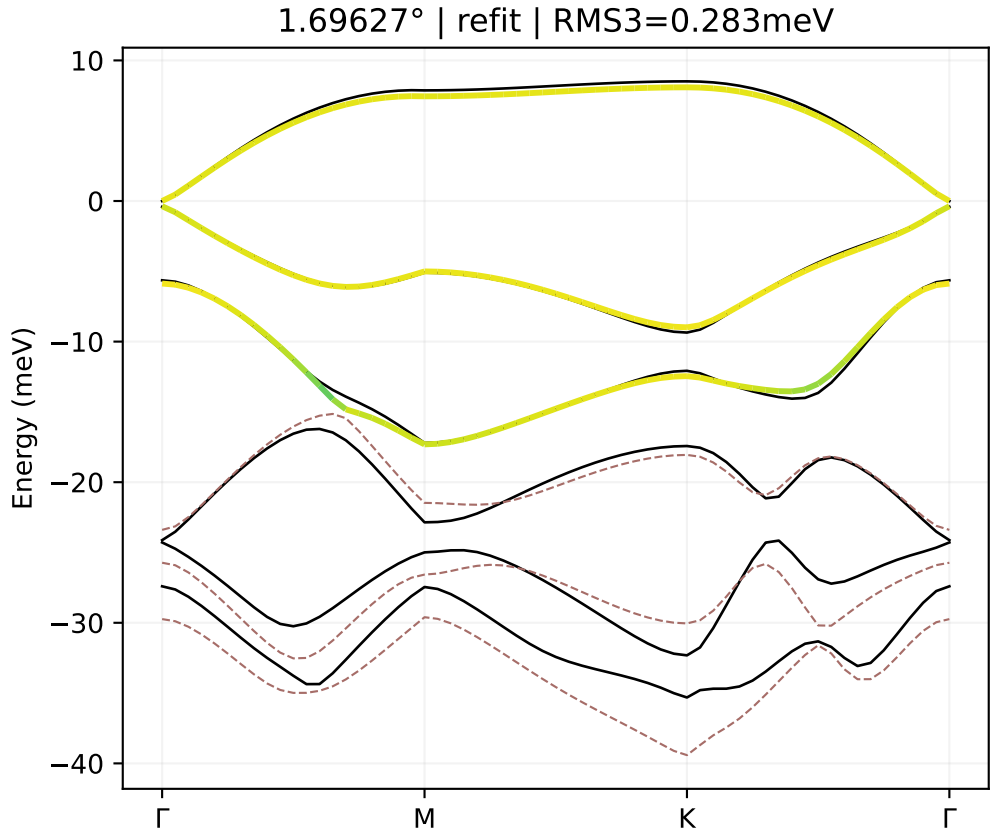
1.61354° | refit | RMS3=0.289meV



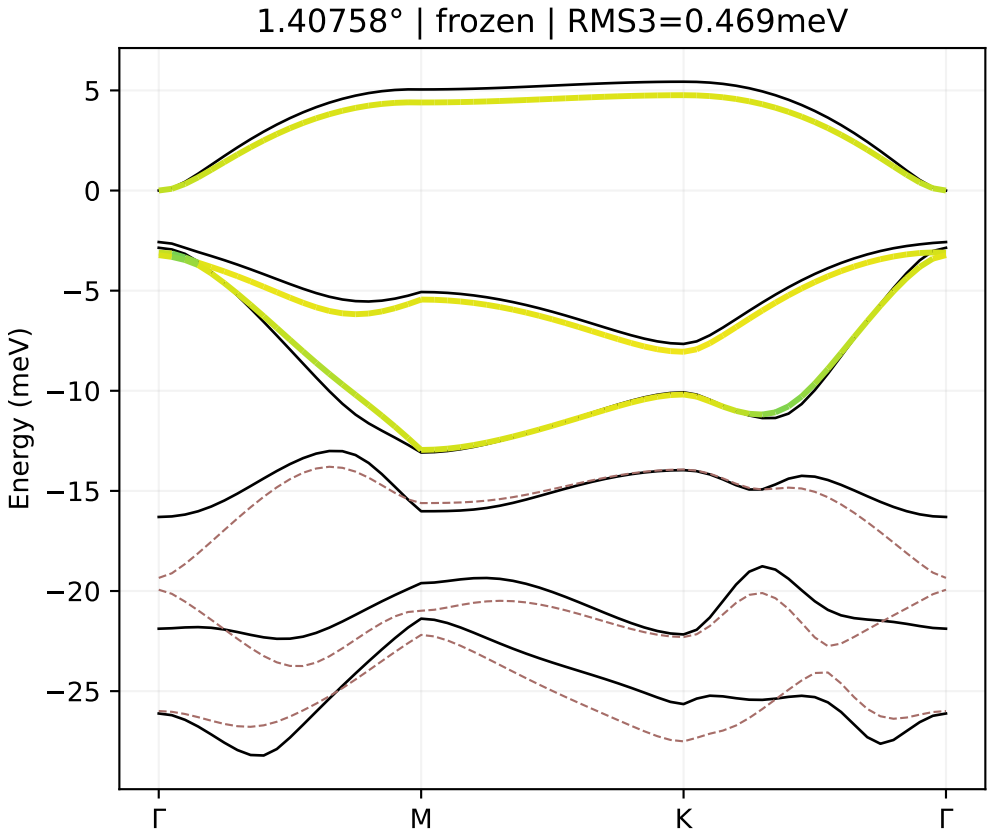
1.69627° | frozen | RMS3=0.320meV



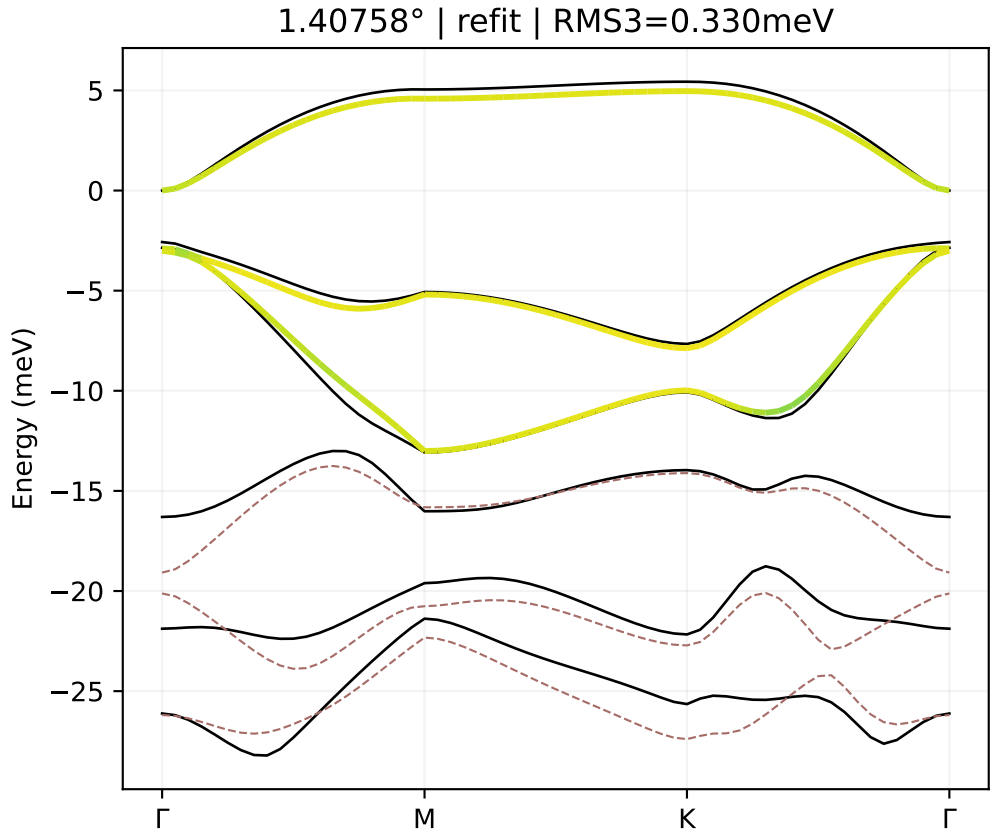
1.69627° | refit | RMS3=0.283meV



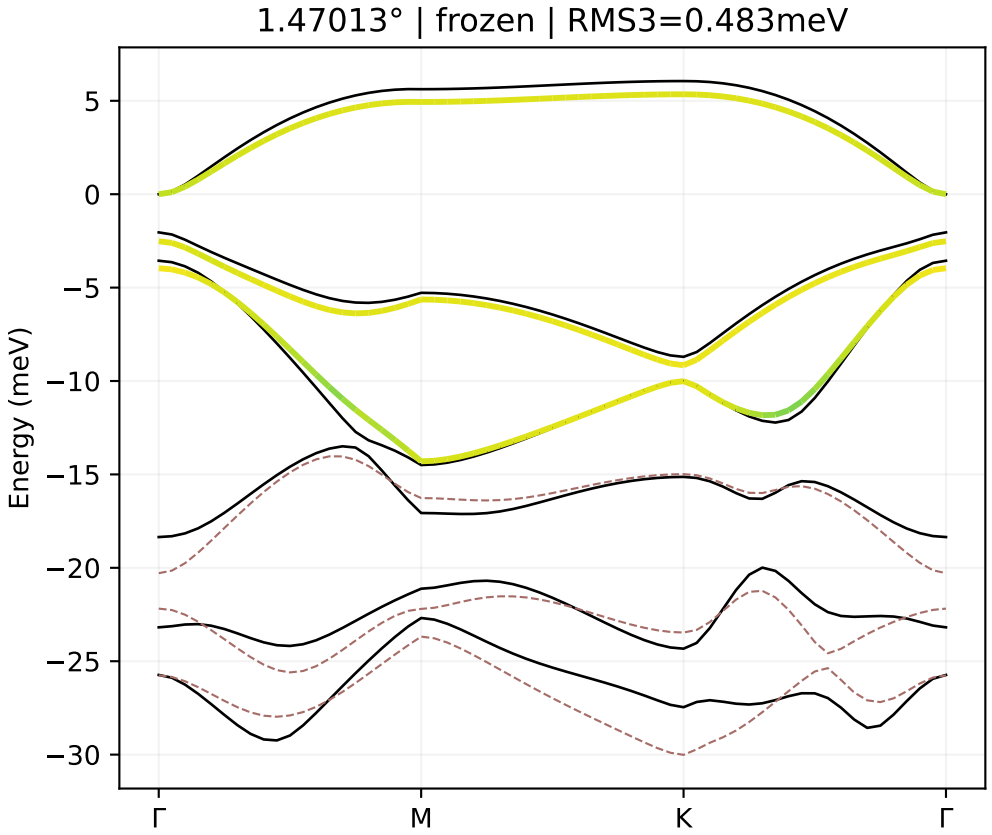
1.40758° | frozen | RMS3=0.469meV



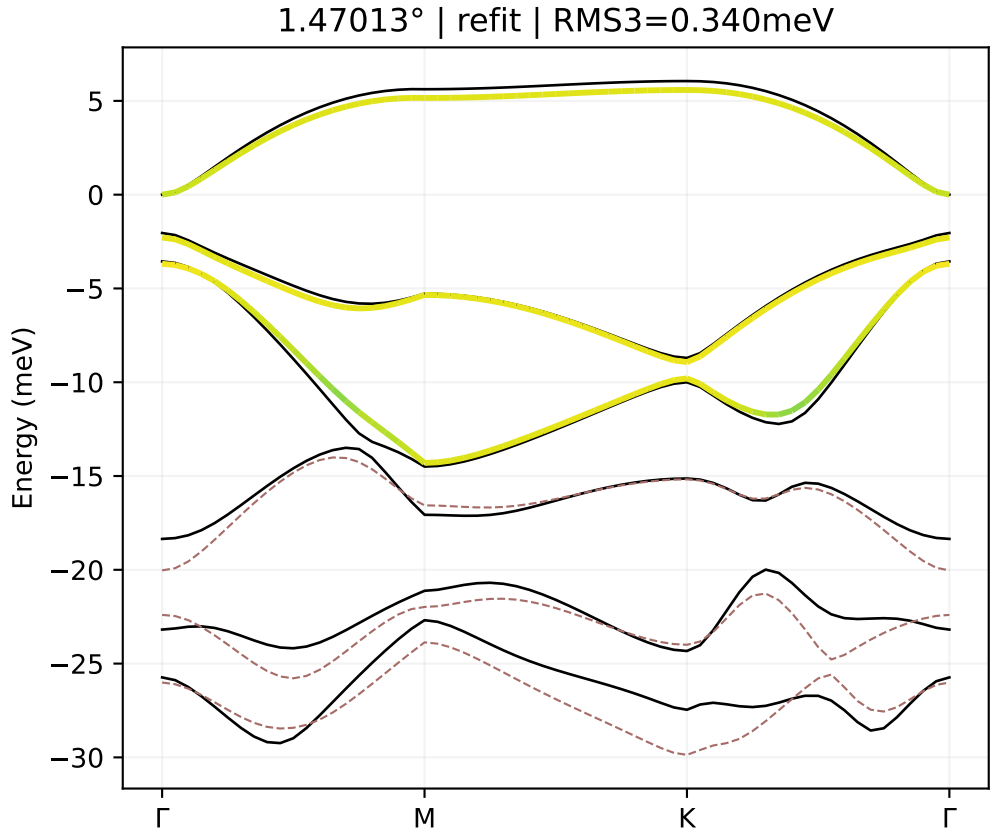
1.40758° | refit | RMS3=0.330meV

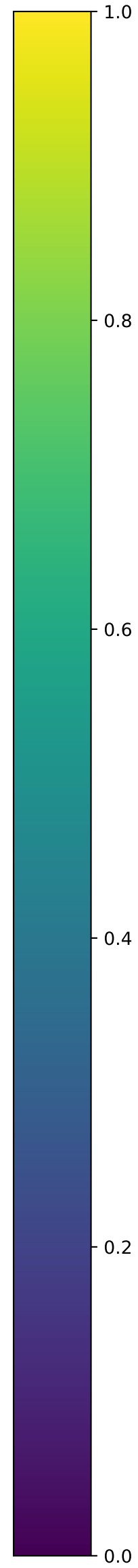


1.47013° | frozen | RMS3=0.483meV

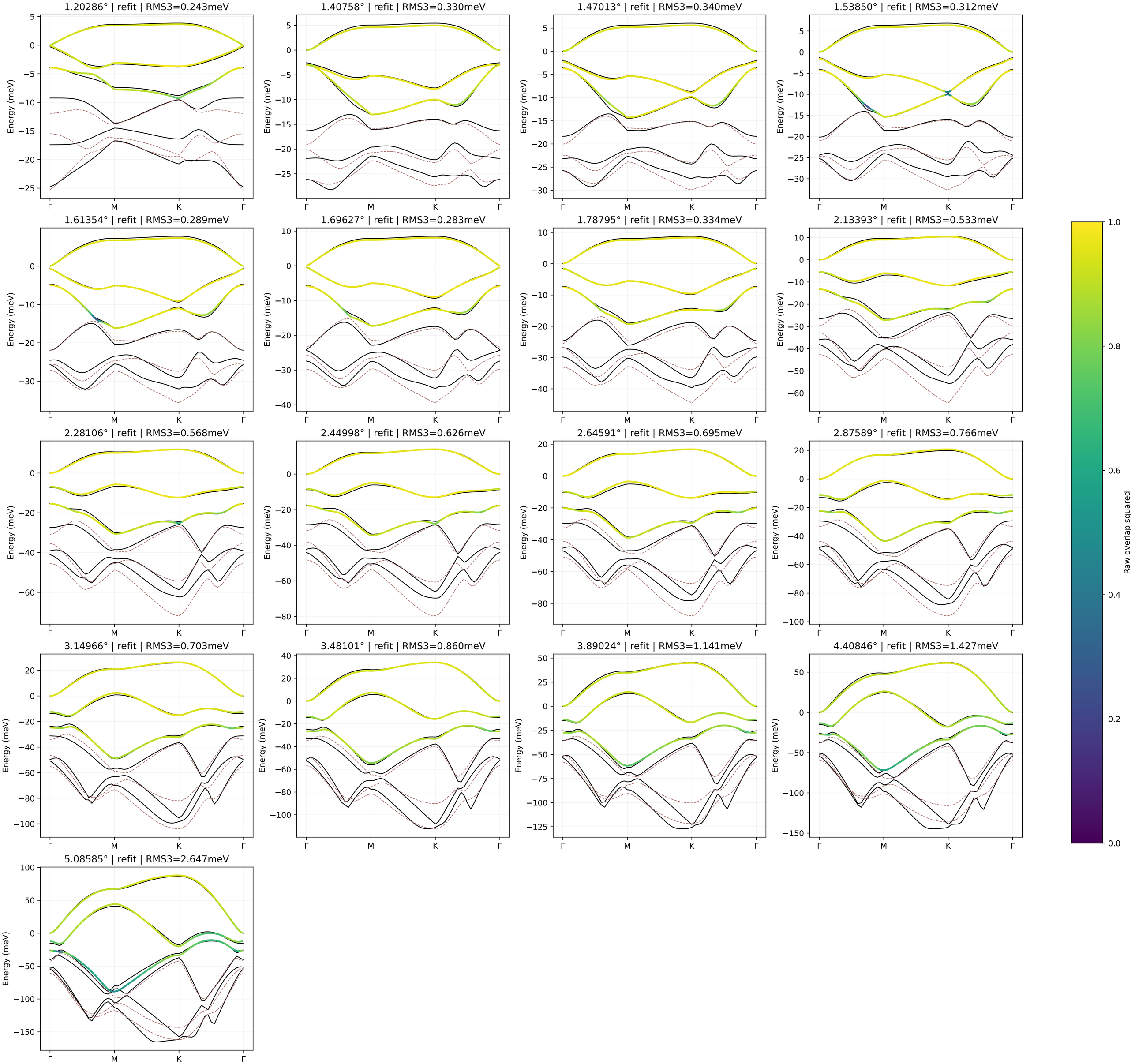


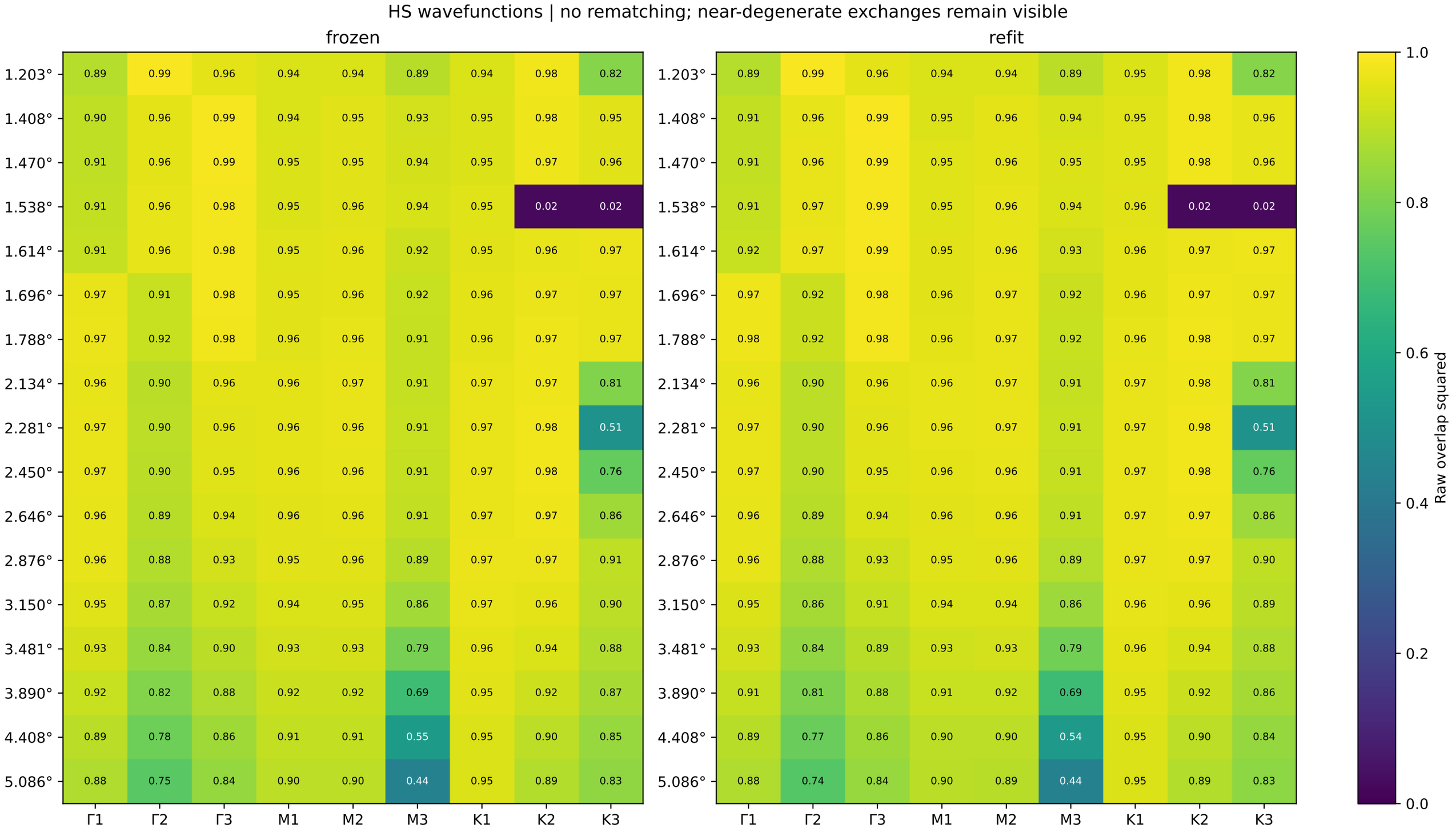
1.47013° | refit | RMS3=0.340meV

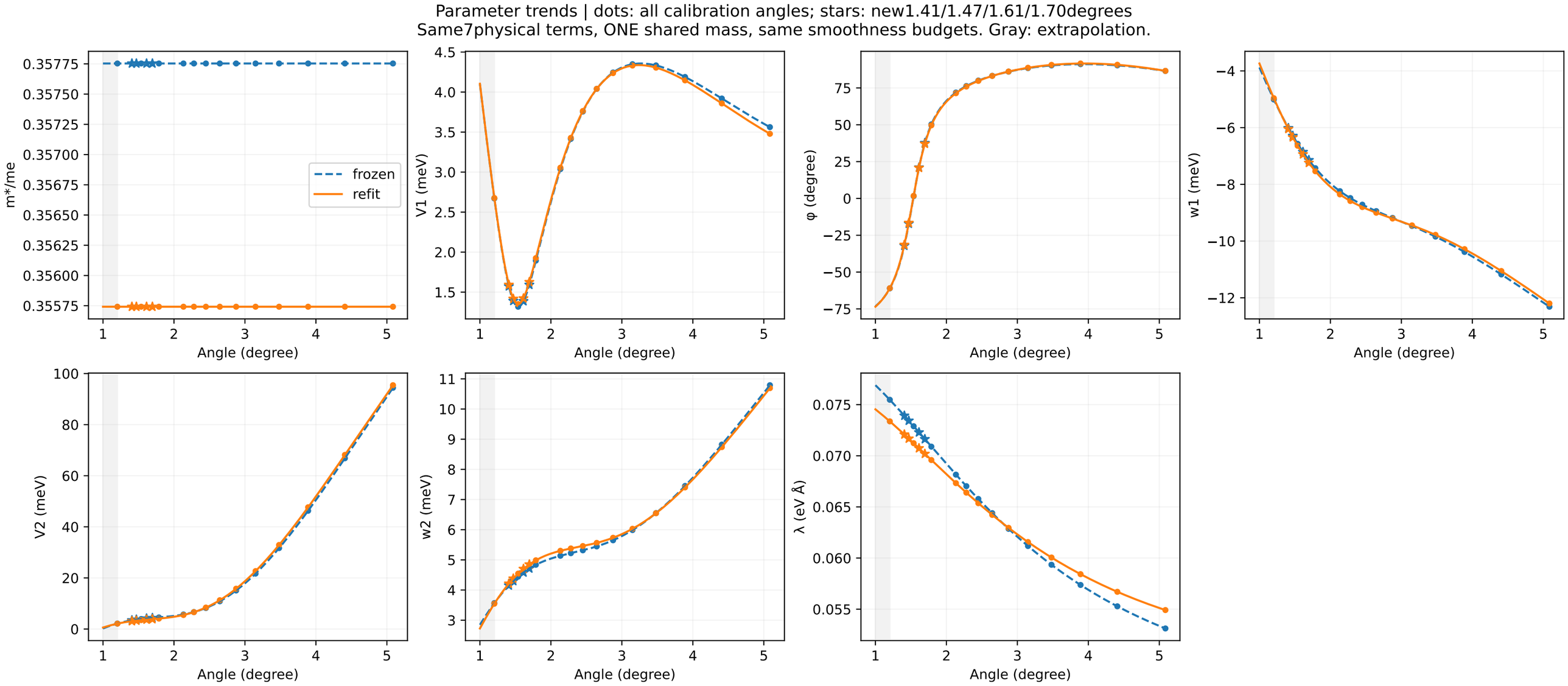


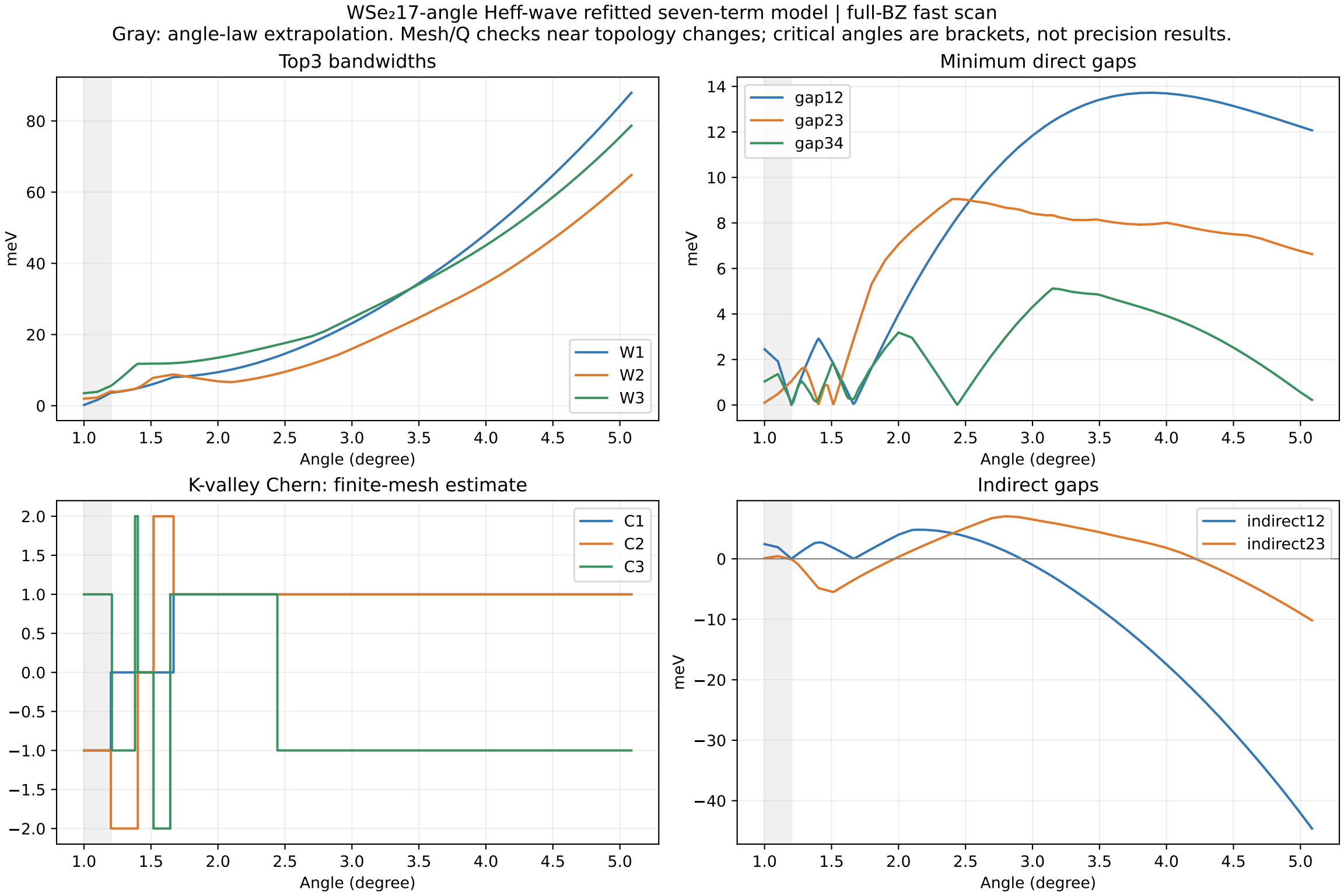


WSe₂17-angle joint refit | top6 bands, unmodified Heff reference




HS wavefunctions | no rematching; near-degenerate exchanges remain visible
frozen
refit
1.203° 0.89 0.99 0.96 0.94 0.94 0.89 0.94 0.98 0.82
1.408° 0.90 0.96 0.99 0.94 0.95 0.93 0.95 0.98 0.95
1.470° 0.91 0.96 0.99 0.95 0.95 0.94 0.95 0.97 0.96
1.538° 0.91 0.96 0.98 0.95 0.96 0.94 0.95 0.02 0.02
1.614° 0.91 0.96 0.98 0.95 0.96 0.92 0.95 0.96 0.97
1.696° 0.97 0.91 0.98 0.95 0.96 0.92 0.96 0.97 0.97
1.788° 0.97 0.92 0.98 0.96 0.96 0.91 0.96 0.97 0.97
2.134° 0.96 0.90 0.96 0.96 0.97 0.91 0.97 0.97 0.81
2.281° 0.97 0.90 0.96 0.96 0.96 0.91 0.97 0.98 0.51
2.450° 0.97 0.90 0.95 0.96 0.96 0.91 0.97 0.98 0.76
2.646° 0.96 0.89 0.94 0.96 0.96 0.91 0.97 0.97 0.86
2.876° 0.96 0.88 0.93 0.95 0.96 0.89 0.97 0.97 0.91
3.150° 0.95 0.87 0.92 0.94 0.95 0.86 0.97 0.96 0.90
3.481° 0.93 0.84 0.90 0.93 0.93 0.79 0.96 0.94 0.88
3.890° 0.92 0.82 0.88 0.92 0.92 0.69 0.95 0.92 0.87
4.408° 0.89 0.78 0.86 0.91 0.91 0.55 0.95 0.90 0.85
5.086° 0.88 0.75 0.84 0.90 0.90 0.44 0.95 0.89 0.83
Γ1 Γ2 Γ3 M1 M2 M3 K1 K2 K3
1.203° 0.89 0.99 0.96 0.94 0.94 0.89 0.95 0.98 0.82
1.408° 0.91 0.96 0.99 0.95 0.96 0.94 0.95 0.98 0.96
1.470° 0.91 0.96 0.99 0.95 0.96 0.95 0.95 0.98 0.96
1.538° 0.91 0.97 0.99 0.95 0.96 0.94 0.96 0.02 0.02
1.614° 0.92 0.97 0.99 0.95 0.96 0.93 0.96 0.97 0.97
1.696° 0.97 0.92 0.98 0.96 0.97 0.92 0.96 0.97 0.97
1.788° 0.98 0.92 0.98 0.96 0.97 0.92 0.96 0.98 0.97
2.134° 0.96 0.90 0.96 0.96 0.97 0.91 0.97 0.98 0.81
2.281° 0.97 0.90 0.96 0.96 0.97 0.91 0.97 0.98 0.51
2.450° 0.97 0.90 0.95 0.96 0.96 0.91 0.97 0.98 0.76
2.646° 0.96 0.89 0.94 0.96 0.96 0.91 0.97 0.97 0.86
2.876° 0.96 0.88 0.93 0.95 0.96 0.89 0.97 0.97 0.90
3.150° 0.95 0.86 0.91 0.94 0.94 0.86 0.96 0.96 0.89
3.481° 0.93 0.84 0.89 0.93 0.93 0.79 0.96 0.94 0.88
3.890° 0.91 0.81 0.88 0.91 0.92 0.69 0.95 0.92 0.86
4.408° 0.89 0.77 0.86 0.90 0.90 0.54 0.95 0.90 0.84
5.086° 0.88 0.74 0.84 0.90 0.89 0.44 0.95 0.89 0.83
Γ1 Γ2 Γ3 M1 M2 M3 K1 K2 K3
Raw overlap squared
1.0
0.8
0.6
0.4
0.2
0.0


Parameter trends | dots: all calibration angles; stars: new1.41/1.47/1.61/1.70degrees
Same7physical terms, ONE shared mass, same smoothness budgets. Gray: extrapolation.
m*/me
frozen
refit
V1 (meV)
φ (degree)
w1 (meV)
V2 (meV)
w2 (meV)
λ (eV Å)
Angle (degree)


WSe₂17-angle Heff-wave refitted seven-term model | full-BZ fast scan
Gray: angle-law extrapolation. Mesh/Q checks near topology changes; critical angles are brackets, not precision results.
Top3 bandwidths
meV
Angle (degree)
W1
W2
W3
Minimum direct gaps
gap12
gap23
gap34
K-valley Chern: finite-mesh estimate
C1
C2
C3
Indirect gaps
indirect12
indirect23